
Chemotherapy

Chemotherapy (chemo) is one of the most common treatments for cancer. Learning about how it works and what to expect can help you prepare for treatment and make informed decisions about your care.

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What is chemotherapy?

Chemotherapy is a treatment that uses medicines to destroy cancer cells. There are many different types of chemo. They don't all work exactly the same way, so different types of chemo might be used for different types of cancer. Most are given as an infusion into a vein (IV), but some are given as an injection, taken as pills, or applied to the skin.

[Watch this video on YouTube¹](#)

What is the goal of chemo?

If your doctor has recommended chemo, it's important to understand the goal of treatment. This can depend on your type of cancer, its stage (size and location), and how far it has spread. Chemo can have 3 possible treatment goals:

To cure the cancer (curative intent)

Whenever possible, the goal of chemo is to cure the cancer – destroying it completely so it doesn't come back. Doctors may call this "treatment with curative intent." They may avoid saying "cure" because it can take many years to know for sure if the cancer is truly gone. Cure is the hope, but it's not always possible.

To control the cancer

If a cure is not possible, chemo may be used to slow the cancer down or shrink it. This can help people feel better and live longer. The [cancer might not go away completely](#)², but it can often be controlled and managed as a chronic disease like heart disease or diabetes. Sometimes the cancer may go away for a while but come back later.

To ease symptoms (palliative intent)

Chemo might be given to improve comfort and quality of life, even when the cancer has spread or can't be controlled. For example, it may help shrink a tumor that's causing pressure or pain. This is called **palliative chemo**.

Palliative chemo is just one part of [palliative care](#)³, which includes things like pain relief and anti-nausea medicine. Palliative treatments can be used at any stage of cancer.

How is chemo different from other cancer treatments?

Chemo is a **systemic treatment**. It travels through the bloodstream to reach all parts of your body. It can kill cancer cells that have spread (metastasized) to parts of the body far away from the original (primary) tumor or cancer cells in blood cancers, such as leukemia, that have spread throughout the body. This makes chemo different from **local treatments** like surgery and radiation, which only affect one part of the body.

How does chemotherapy work?

Understanding how chemotherapy works can help you know what to expect during treatment.

Chemotherapy interrupts the cell cycle

Chemotherapy works by affecting cells when they are growing and dividing to make new cells. During this process (called the **cell cycle**), cells:

- Grow in size
- Make copies of their genetic material (DNA)
- Divide to form new cells

Some cells, such as skin cells, are fast growing, meaning they move through this process quickly. Other cells, such as muscle cells, complete it more slowly. Cancer cells tend to be faster growing, moving through the cell cycle very quickly.

Different types of chemo target cells at different phases of the cell cycle. This is why certain chemo drugs work better for different types of cancer cells, and sometimes different combinations of chemo work better together. Understanding how chemo works also helps doctors plan how often each chemo should be given, and how those doses should be timed.

Chemo affects normal cells, too

Chemo interrupts the cell cycles of normal cells, too – especially cells that grow faster, like hair or skin cells. This is why chemo often causes certain side effects, like hair loss.

The good news is that most normal cells will recover from chemo. Chemo is a useful cancer treatment because cancer cells are abnormal (mutated) and less able to recover from its effects.

To work best, chemo treatments must be at just the right dose to kill the cancer cells while sparing as many normal cells as possible.

[What Goes into Planning Your Chemotherapy](#) [Types of Chemotherapy](#)
[Chemo Safety](#) [Chemo Side Effects](#)

Are all medicines used to treat cancer called chemo?

Not all cancer medicines are chemotherapy. Other medicines and biological treatments may be used alone or with chemo or other treatments. They work differently and often have different side effects.

- [Targeted therapy](#)⁴: These find and attack specific proteins or receptors that some cancer cells have, without affecting most normal cells.
- [Hormone therapy](#)⁵: These block or lower hormones that help some cancers (like breast, prostate, and endometrial (uterine) cancers) grow. They either stop the body from making the hormone or block the cancer cells from using it.
- [Immunotherapy](#)⁶: These help a person's immune system recognize and attack certain types of cancer cells.

Although these treatments can be very effective, they are more specialized than traditional chemo. They might only work on certain types of cancer, often based on specific traits of the cancer cells. And because many of them are newer, less is known about their long-term side effects. Sometimes they are used only when chemo isn't working or is no longer effective.

What is it like to get chemo?

Most people get chemo as infusions at a clinic or hospital. But some types of chemo can be taken by mouth or applied to the skin at home. Each type has its own routine, side effects, and safety steps. Learn what it's like to get each type of chemo and how to stay safe during treatment.

[Chemo Infusions and Injections](#)

Chemo is most often given as an infusion or injection into a vein, but there are other ways to do chemo infusions as well.

[Oral Chemotherapy](#)

Oral chemotherapy is taken by mouth as pills, capsules, or liquids.

[Topical Chemotherapy](#)

Topical chemo comes in a cream or ointment.

Learn about chemotherapy for your cancer type

Choose a cancer type Acute Lymphocytic Leukemia (ALL) Acute Myeloid Leukemia (AML) Adrenal

Cancer Anal Cancer Basal and Squamous Cell Skin Cancer Bile Duct Cancer Bladder Cancer Bone Cancer Brain and Spinal Cord Tumors Brain and Spinal Cord Tumors in Children Breast Cancer Cancer of Unknown Primary Cervical Cancer Childhood Non-Hodgkin Lymphoma Chronic Lymphocytic Leukemia (CLL) Chronic Myelomonocytic Leukemia (CMML) Colon and Rectal Cancer Endometrial Cancer Esophageal Cancer Ewing Tumors Eye Cancer Gallbladder Cancer Gastrointestinal Carcinoid Tumors Gastrointestinal Carcinoid Tumors Gastrointestinal Stromal Tumor (GIST) Hodgkin Lymphoma Kaposi Sarcoma Kidney Cancer Laryngeal and Hypopharyngeal Cancer Leukemia in Children Liver Cancer Lung Carcinoid Tumor Melanoma Merkel Cell Skin Cancer Mesothelioma Multiple Myeloma Myelodysplastic Syndrome Nasal Cavity and Paranasal Sinus Cancer Nasopharyngeal Cancer Neuroblastoma Non-Hodgkin Lymphoma Non-small Cell Lung Cancer Oral Cavity and Oropharyngeal Cancer Osteosarcoma Ovarian Cancer Pancreatic Cancer Pancreatic Neuroendocrine Tumor (NET) Penile Cancer Prostate Cancer Retinoblastoma Rhabdomyosarcoma Salivary Gland Cancer Small Cell Lung Cancer Small Intestine Cancer Soft Tissue Sarcoma Stomach Cancer Testicular Cancer Testicular Cancer Thymus Cancer Thyroid Cancer Uterine Sarcoma Vaginal Cancer Vulvar Cancer Waldenstrom Macroglobulinemia Wilms Tumor

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Hyperlinks

1. www.youtube.com/embed/gnIGpuSWRIQ?origin=http%3A%2F%2Flocalhost%3A4502&hl=en&enablejsapi=1
2. www.cancer.org/cancer/survivorship/long-term-health-concerns/cancer-as-a-chronic-illness.html
3. www.cancer.org/cancer/supportive-care/palliative-care.html
4. www.cancer.org/cancer/managing-cancer/treatment-types/targeted-therapy.html
5. www.cancer.org/cancer/managing-cancer/treatment-types/hormone-therapy.html
6. www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy.html

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What Goes into Planning Your Chemotherapy

Planning chemotherapy involves more than just picking a medication. Your cancer care team considers many factors—like cancer type, treatment goals, and your overall health—to decide which type of chemo to use, how much to give, and how often. Understanding how these decisions are made can help you feel more informed and prepared.

[How chemo works with other treatments](#)

Choosing the right chemo
Choosing the right chemo dose
Chemo cycles
When treatment plans change

How chemo works with other treatments

Sometimes chemo is the only treatment you need (primary treatment). Or, it might be used along with other treatments. For example, chemo may be used:

- To shrink a tumor before surgery or radiation therapy. Chemo used in this way is called **neoadjuvant therapy**.
- After surgery or radiation therapy to help kill any remaining cancer cells in the body. Chemo used in this way is called **adjuvant therapy**.
- With other types of medicines, such as targeted therapy that acts on certain targets of cancer cells or immunotherapy that helps the immune system fight cancer.
- With other treatments if cancer comes back or doesn't completely go away.

Choosing the right chemo

Cancer is sometimes treated with a single chemo drug, but often, several drugs are used together. They may be given in a certain order or in certain combinations (called **combination chemo**). Using 2 or more drugs that work in different ways can kill more cancer cells than using one by itself. It can also help lower the chance that the cancer becomes resistant to any one type of chemo.

In some cases, the best doses and schedules for each drug are well known, based on research studies in people with similar cancers.

In other cases, less may be known about the best way to treat a certain type or stage of cancer. Your overall health, other medical conditions, or medicines you take might also affect your treatment options. In these cases, different doctors might choose different drug combinations or schedules.

Factors a cancer care team considers when recommending treatment options include:

- The type and subtype of cancer

- The stage of the cancer (where it is, how big it is, and whether it's spread to other parts of the body)
- Results of other tests on the tumor, such as biomarkers
- Your age
- Your overall health and current medications
- Other serious health problems (such as heart, liver, or kidney diseases)
- Goal of treatment
- Types of cancer treatments you have had in the past

If your treatment plan is not clear, or you are given several possible options, ask your cancer care team about [getting a second opinion](#)¹. It's important to feel confident that you understand your choices and are making the decision that's right for you.

Choosing the right chemo dose

Doctors must calculate chemo doses very carefully because most chemo drugs are strong medicines with a narrow dose range for safety and effectiveness. Too little may not treat the cancer, and too much can cause life-threatening side effects.

There are different ways to determine chemo dosage (amount), depending on the drug. Most chemo drugs are measured in milligrams (mg).

Chemo dosage by body weight

The overall dose may be based on a person's **body weight in kilograms** (1 kilogram is 2.2 pounds).

For example, if 10 milligrams (mg) of a drug is needed for each kilogram (kg) of body weight, a person weighing 50 kilograms (110 pounds) would get 500 milligrams of the drug ($50\text{kg} \times 10\text{mg/kg} = 500\text{mg}$).

Chemo dosage by body surface area

Some chemo doses are determined based on **body surface area (BSA)**, which is calculated using height and weight. BSA is expressed in meters squared (m^2).

Chemo dosage for children

Because children's bodies process medicines differently, dosages for children and adults differ, even after BSA is taken into account. Children may have different levels of sensitivity to the chemo, too.

Other factors that can affect chemo dosage

Dosages of some chemo may also be adjusted for people who:

- Are over age 65
- Have poor nutrition
- Have already taken or are currently taking other medicines
- Have already had or are currently getting radiation therapy
- Have low blood cell counts
- Have liver or kidney diseases
- May otherwise be unable to tolerate full doses

Chemo cycles

Chemo is usually given at regular intervals called **cycles**. A cycle may be a dose of one or more types of chemo on one or more days, followed by several days or weeks without treatment. This gives normal cells time to recover from the effects of the chemo.

Some drugs are given for several days in a row, or every other day, followed by a rest period. Others work best when given continuously over a set number of days.

Each drug is given on a schedule that makes the most of its anti-cancer actions and minimizes side effects. If more than one type of chemo is used, the treatment plan will say how often and exactly when each should be given.

The number of cycles given may be decided before treatment starts, based on the type and stage of cancer. In some cases, the number is flexible and will take into account how the treatment affects the cancer and the person's overall health.

When treatment plans change

In most cases, the most effective chemo doses and schedules for specific cancers have been found by testing them in [clinical trials](#)². It's important, when possible, to get the full course of chemo, the full dose, and keep the cycles on schedule. This gives you the best chance of getting the maximum benefit from treatment.

There may be times, though, when serious side effects require adjusting the chemo plan (dose and/or schedule) to allow your body time to recover. Sometimes, you might be given supportive medicines to help you recover more quickly. Or sometimes, the cancer cells develop resistance to the chemo and a different type might be needed.

The goal is always to give enough chemo to treat the cancer effectively, while keeping side effects as manageable as possible. Your cancer care team will work with you to adjust the plan as needed and support you through treatment.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/seeking-a-second-opinion.html
2. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/clinical-trials.html

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Types of Chemotherapy Drugs

Understanding the chemotherapy drugs you're getting can help you feel more informed and prepared. Knowing the group that the chemo belongs to—and its exact name—can give you helpful context for why certain drugs are used, combined, or even substituted during treatment.

What are chemo groups?

What are chemo groups?

Chemo drugs can be grouped by how they work, their chemical structure, and their relationships to other types of chemo. Some drugs work in more than one way and may belong to more than one group or class. These groupings help doctors choose drugs that:

- Target cancer in different ways
- Work well together
- Have manageable side effects

Ask your cancer care team to explain what kind of chemo you are getting, how it works, and, if you are getting more than 1 type, how they work together.

Your doctor may recommend substituting one chemo for another in the same class. This may happen if you're having side effects or if there is a shortage of a certain drug.

Also, sometimes cancer cells develop resistance to a particular chemo causing it to stop working. If this happens, your doctor may recommend changing to another chemo. This may be a drug within the same class that works slightly differently, or it may be one from another class. Changing drugs can help kill cancer cells that no longer respond to the old drug and help prevent new resistance from happening.

Why exact drug names matter

Chemo medicines can have various names, including brand names and generic names. They may also be referred to by an abbreviation or as part of an acronym used for a specific chemo combination (regimen). This can be confusing, so ask your cancer care team to write down exactly what medications you'll be getting.

Even when 2 types of chemo are in the same group or have similar names, they can act very differently in the body. Be sure you know the exact, full name of your chemo medicines when looking up side effects or other information.

Alkylating agents (including platinum drugs)

How they work: Alkylating agents keep the cell from reproducing (making copies of itself) by damaging its DNA. These work in all phases of the cell cycle. While not true alkylating agents, platinum drugs (carboplatin, cisplatin, oxaliplatin) are often classified as alkylating agents because they also work by damaging cancer cell DNA, just in a different way.

Cancers treated: Alkylating agents are used to treat many different cancers, including cancers of the lung, breast, and ovary, as well as leukemia, lymphoma, Hodgkin disease, multiple myeloma, and sarcoma.

Side effects: Because these drugs damage DNA, they can affect the cells of the bone marrow, which make new blood cells. In rare cases, this can lead to [myelodysplastic syndrome](#)¹ or acute [leukemia](#)². The risk of leukemia from alkylating agents is “dose-dependent,” meaning that the risk is small with lower doses, but goes up as the total amount of the chemo used gets higher. The risk of leukemia after getting alkylating agents is highest about 5 to 10 years after treatment.

Examples of alkylating agents include:

- Bendamustine
- Busulfan
- Carboplatin
- Chlorambucil
- Cisplatin
- Cyclophosphamide
- Dacarbazine
- Ifosfamide
- Mechlorethamine
- Melphalan
- Oxaliplatin
- Temozolomide
- Thiotepa
- Trabectedin

Nitrosoureas

Nitrosoureas are a group of alkylating agents that are able to cross through the **blood-brain barrier** to reach cancer cells in the brain. Most other chemo types, including other alkylating agents, cannot cross the blood-brain barrier.

Cancers treated: These are useful in treating certain types of brain tumors.

Examples of nitrosoureas include:

- Carmustine
- Lomustine
- Streptozocin

Antimetabolites

How they work: Antimetabolites interfere with DNA and RNA by acting as a substitute for the normal building blocks of RNA and DNA. When this happens, the DNA cannot make copies of itself (replicate), and a cell cannot reproduce.

Cancers treated: Antimetabolites are commonly used to treat leukemias, cancers of the breast, ovary, and the intestinal tract, as well as other types of cancer.

Examples of antimetabolites include:

- Azacitidine
- 5-fluorouracil (5-FU)
- 6-mercaptopurine (6-MP)
- Capecitabine
- Cladribine
- Clofarabine
- Cytarabine
- Decitabine
- Floxuridine
- Fludarabine
- Gemcitabine
- Hydroxyurea
- Methotrexate
- Nelarabine
- Pemetrexed
- Pentostatin
- Pralatrexate
- Thioguanine

Anti-tumor antibiotics

How they work: Anti-tumor antibiotics are not like the antibiotics used to treat infections. They work by changing the DNA inside cancer cells to keep them from growing and multiplying.

Anthracyclines

Anthracyclines are anti-tumor antibiotics that interfere with enzymes involved in copying DNA during the cell cycle. They bind with DNA so it cannot make copies of itself, and a cell cannot reproduce. (Enzymes are proteins that start, help, or speed up the rate of chemical reactions in cells.) They are widely used for a variety of cancers.

Examples of anthracyclines include:

- Daunorubicin
- Doxorubicin
- Doxorubicin liposomal
- Epirubicin
- Idarubicin
- Valrubicin

Side effects: A major concern when giving these drugs is that they can permanently damage the heart if given in high doses. For this reason, lifetime dose limits (also called **cumulative dose**) are often placed on these medicines.

Other anti-tumor antibiotics

Anti-tumor antibiotics that are not anthracyclines include:

- Bleomycin
- Dactinomycin
- Mitomycin-C
- Mitoxantrone (also acts as a topoisomerase II inhibitor, see below)

Topoisomerase inhibitors (plant alkaloids)

Topoisomerase inhibitors are also called **plant alkaloids**.

How they work: These interfere with enzymes called **topoisomerases**, which help separate the strands of DNA so they can be copied. (Enzymes are proteins that cause chemical reactions in living cells.)

Cancers treated: Topoisomerase inhibitors are used to treat certain leukemias, as well as lung, ovarian, gastrointestinal, colorectal, and pancreatic cancers.

Topoisomerase inhibitors are grouped according to which type of enzyme they affect:

Topoisomerase I inhibitors (also called **camptothecins**) include:

- Irinotecan
- Irinotecan liposomal
- Topotecan

Topoisomerase II inhibitors (also called **epipodophyllotoxins**) include:

- Etoposide
- Mitoxantrone (also acts as an **anti-tumor antibiotic**)
- Teniposide

Side effects: Topoisomerase II inhibitors can increase the risk of a [second cancer](#)³.

Mitotic inhibitors (taxanes and vinca alkaloids)

Mitotic inhibitors are another type of plant alkaloid. They include taxanes and vinca alkaloids. They are compounds derived from natural products, such as plants.

How they work: These work by stopping cells from dividing to form new cells. They can damage cells in all phases by keeping enzymes from making proteins needed for cell reproduction.

Cancers treated: Mitotic inhibitors are used to treat many different types of cancer including breast, lung, myelomas, lymphomas, and leukemias.

Side effects: These may cause [nerve damage](#)⁴, which can limit the amount that can be given.

Examples of mitotic inhibitors include the taxanes and vinca alkaloids.

Taxanes include:

- Cabazitaxel

- Docetaxel
- Nab-paclitaxel
- Paclitaxel

Vinca alkaloids include:

- Vinblastine
- Vincristine
- Vincristine liposomal
- Vinorelbine

Corticosteroids

Corticosteroids, often simply called **steroids**, are natural hormones and hormone-like medicines that are useful in the treatment of many types of cancer, as well as other illnesses. When used as part of cancer treatment, they are considered chemo.

Examples of corticosteroids include:

- Prednisone
- Methylprednisolone
- Dexamethasone

Steroids are also commonly used to help prevent nausea and vomiting caused by chemo. They are used before some types of chemo to help prevent severe allergic reactions, too.

Other types of chemotherapy

Some chemo drugs act in slightly different ways and do not fit well into any of the other categories. Here are some examples:

- All-trans-retinoic acid
- Arsenic trioxide
- Asparaginase
- Eribulin
- Hydroxyurea

- Ixabepilone
- Mitotane
- Omacetaxine
- Pegaspargase
- Procarbazine
- Romidepsin
- Vorinostat

Learn about chemotherapy for your cancer type

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Hyperlinks

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2. www.cancer.org/cancer/types/leukemia.html
3. www.cancer.org/cancer/survivorship/long-term-health-concerns/second-cancers-in-adults.html
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Getting Chemo Infusions or Injections

Many types of chemotherapy (chemo) are given as an infusion or injection. The most common way is through a thin tube (**catheter**) placed in a vein. Chemo can also be delivered into an artery, body cavity, or body part through a catheter, or injected quickly through the skin using a needle and syringe.

[What is intravenous \(IV\) chemo?](#)

[Will I need a port?](#)

[Where will I get chemo?](#)

[How should I prepare for chemo?](#)

[What will chemo infusions be like?](#)

[Other ways chemo is given](#)

[How often and for how long will I need chemo?](#)

[How long does chemo stay in the body?](#)

What is intravenous (IV) chemo?

Intravenous (IV) chemo is put right into your bloodstream. Often, this is done through a tiny, soft, plastic tube called an **IV catheter**. A needle is used to put the catheter into a vein in your forearm or hand. Then the needle is taken out, leaving the catheter behind.

Once the IV is in place, chemo and other medicines can be given in several ways, depending on the type:

- **IV push:** The medicines can be given quickly through the catheter right from a syringe over a few minutes.
- **IV infusion:** A typical infusion can last from a few minutes to a few hours. A mixed solution containing the chemo flows from a plastic bag through tubing that's attached to the catheter. The flow is usually controlled by a machine called an **IV pump**.
- **Continuous infusion:** These infusions can last anywhere from 1 to several days and are controlled by electronic IV pumps.

Will I need a port?

Some people may get a new IV catheter placed in their hand or arm for each chemo infusion. But over time, this can damage your veins. It can become harder to find a good vein after many treatments and blood draws for tests.

If you will need chemo for an extended period of time, your cancer care team might suggest placing a **port**, also called a **central venous catheter** or **CVC**. A CVC is a bigger catheter that's put into a large vein in the chest or arm. It stays in as long as you're getting treatment so you won't need to be stuck with a needle each time you need treatment or tests.

Usually, placing a port is done with a minor surgical procedure, in a clinic, hospital room, or operating room.

Once your port is in place, you:

- Can get all your treatments through the port
- Will not be stuck multiple times for IVs
- Can have most blood tests drawn through it
- Can get other treatments and medicines you might need through it, such as fluids, blood transfusions, or antibiotics

Many people talk about CVC options with their doctor before their first chemo treatment. Your health care team can help you decide if you need a CVC and what type is right for you. Even if you don't get a port before starting treatment, you can choose to get one later if it becomes harder to find a usable vein for infusions, injections, or blood draws.

Learn more about the [different types of ports and CVCs¹](#).

Where will I get chemo?

Where you get your chemo depends on:

- The type of chemo and its dosage
- Hospital policies
- Insurance coverage
- Your preferences

- Your doctor's recommendation

You may get chemo:

- At home
- In your doctor's office
- In a clinic
- In a hospital's outpatient infusion center
- In a hospital

Some places have private treatment rooms, while others treat many patients together in one large room. Ask your doctor or nurse about this ahead of time so you know what to expect on your first day.

How should I prepare for chemo?

Your cancer care team will give you detailed instructions about what to do before your infusion day. **Depending on the type of cancer you have, the type of chemo you will receive, and the possible side effects**, they might suggest that you:

- Start certain medicines or supplements
- Get ice boots and mittens or compression socks and gloves to wear during your infusion to limit the effects of chemo in your hands and feet
- Consider whether you want to use a [cold cap](#)² to help prevent hair loss

You can prepare in other ways, too:

- Make sure you have a ride to and from treatment. If you don't have a friend or caregiver to go with you, call the American Cancer Society. [Road To Recovery](#)³ can match you with a volunteer in your area.
- Plan what you'll wear. Chemo infusions can last several hours or most of the day. Wearing something comfortable can help.
- If you're cold-natured, have a blanket ready to bring with you. Infusion rooms are often chilly.
- If you're likely to lose your hair, think about what you'd like to do. Some people choose to cut their hair short before starting chemo. Some prepare by being [fitted for a wig](#)⁴ or finding a [scarf](#)⁵ or [hat](#)⁶ to wear.

Ask your care team what to expect in the days after your infusion, too. It can help to be prepared with things like easy meals, any medicines they recommend for side effects, and a few comfort items at home. You might want to keep your schedule light for the first few days after treatment, in case you feel tired or need time to rest. Planning ahead can make a big difference in helping you feel more comfortable and in control.

What will chemo infusions be like?

Here's what to expect on the day of your chemo infusion.

Before the infusion

Before your chemo starts, there are a few steps to make sure you're ready and it's safe to move forward with treatment.

- A nurse will access your port or start an IV in your hand or arm. (This is often done at the same time as a blood draw.)
- Your vital signs, height, weight, and blood counts will be checked to make sure it's safe to get chemo.
- You might also see your doctor or nurse practitioner (NP)/physician assistant or associate (PA) for a physical exam.

Getting the infusion

You will be seated in the infusion room in a chair, usually a comfortable recliner made specially for chemo treatments. To get the chemo infusion, your IV or port will be connected to a tube or line.

Depending on the specific chemo you are getting, you may be given other IV medicines first. These are called "pre-meds" and may include:

- Steroids
- Allergy medicines
- Anti-nausea medicines

You also may be given IV fluids ahead of time to make sure you are well hydrated.

Once the pre-meds and fluids are complete, your chemo will be started. Some pre-meds can make you feel sleepy, but you should not feel anything from the chemo medication while it is infusing. Your nurse will check on you throughout the infusion to make sure you are tolerating it well and not having any reaction to it. They will check your IV site and may repeat your vital signs.

While the chemo is infusing, you can watch TV, read a book, or talk with the person who came with you.

How long an infusion lasts will depend on many factors. Some chemo treatments take minutes or hours. Others are given over several days or weeks. This is called **continuous infusion chemo**. You do not need to stay at the hospital or clinic for continuous infusion. Instead, chemo is delivered through a small pump you wear or carry.

After the infusion

When your infusion is done, the nurse will:

- Flush your IV line and remove it (or disconnect it from your port or CVC)
- Check on how you are feeling and may repeat your vital signs
- Give you instructions to follow once you return home
- Make sure you have any prescriptions (like anti-nausea meds) you may need
- Set up appointments for blood work and other follow-ups
- Tell you who to contact if you have any problems after the infusion

Other ways chemo is given

Chemotherapy can also be given in other ways, depending on the type of cancer and where it is in the body. Sometimes it's given into the spinal fluid, arteries, body cavities, or as a shot. In some cases, these methods help target the cancer more directly and may cause fewer side effects.

Intrathecal (IT) chemotherapy

Intrathecal or **IT chemotherapy** is chemo that's given directly into the fluid around your brain and spinal cord (**cerebrospinal fluid** or **CSF**). It may be used for cancers that affect the brain because many chemo drugs given by IV or by mouth cannot cross the blood-brain barrier to reach this area.

IT chemo can be given in two ways:

- **Spinal injection:** A needle is inserted into your lower back to deliver the chemo into the spinal canal.
- **Ommaya reservoir:** A small device is placed under the scalp during surgery, with a tube that goes into a fluid-filled space in the brain. It stays in place until no more treatments are needed.

IT chemo is usually given in a doctor's office, clinic, hospital's outpatient infusion center or radiology department, or in the hospital.

Here's what to expect:

- Your vital signs, height, weight, and blood counts will be checked to make sure it's safe to get chemo.
- If needed, a port or IV may be used to give fluids or medications to help prevent side effects.
- You'll be positioned on a bed, table, or in a chair for the procedure.
- The doctor, NP, or PA will clean your skin and give the chemo.

After the chemo:

- If the chemo was given through a needle into the spinal area, you may be asked to lie flat on the table for a while afterward.
- Your vital signs will be checked again.
- Before going home, you'll get instructions for home care and any prescriptions you might need.

Intra-arterial chemo

Intra-arterial (IA) chemotherapy is chemo that's put directly into the main artery that supplies blood to the tumor. This allows a high dose of chemo to go straight to the tumor, while limiting side effects in the rest of the body. It might be used to treat:

- A single organ, such as the liver
- A limb, such as an arm or leg
- The eye (usually in children)

IA chemo is usually done in a hospital's radiology department or an operating room.

What to expect:

- Your skin will be cleaned and prepped.
- The doctor will make a small cut and insert a catheter (thin tube) through your skin and into an artery near the tumor.
- The chemo is given through the catheter. Sometimes a pump is attached to deliver chemo continuously over a period of time.

After the chemo:

- Once the chemo is done, the catheter is removed and pressure will be applied to help stop any bleeding.
- You will be monitored for a while to check for bleeding, swelling, or poor circulation (such as numbness or coolness).
- The area will be covered with a bandage or dressing.
- Before going home, you'll get instructions for home care and any prescriptions you might need.

Intracavitary chemo

Intracavitary chemotherapy is chemo that's given directly into an enclosed space inside your body. This lets the chemo reach the cancer more directly and helps limit side effects elsewhere. It may be used in areas such as:

- The bladder (**intravesicular** or **intravesical chemo**)
- The abdomen or belly (**intraperitoneal chemo**)
- The chest (**intrapleural chemo**)

Intracavitary chemo is usually done in a clinic, radiology department or hospital operating room.

What to expect:

- Depending on where the cancer is, the chemo is given through a urinary catheter into the bladder, intraperitoneal catheter or port into the abdomen, or chest tube into the lung.
- The chemo stays in the body for a period of time and is then drained out.

After the chemo:

- You will be monitored for a while to check for any side effects or reactions.
- Before going home, you'll get instructions for home care and any prescriptions you might need.

Subcutaneous (SC or SQ) and Intramuscular (IM) chemo

Some types of chemo are given as an injection or shot rather than through an IV. These shots go into specific layers of tissue:

- **Subcutaneous (SC or SQ) chemo**⁷ is injected into the fatty tissue just under the skin.
- **Intramuscular (IM) chemo** is injected deeper into a muscle.

Sometimes more than 1 injection is needed, depending on the amount of chemo being given.

SQ and IM chemo are usually given in a doctor's office, clinic, or hospital's outpatient infusion center.

What to expect:

- The nurse will clean your skin before the injection.
- The needle is inserted into the correct area—just under the skin (SQ) or into the muscle (IM).
- The chemo is injected and the needle is removed.

- A bandage is applied.
- If more injections are required, this process will be repeated.

After the chemo:

- You will be monitored for any bleeding, swelling, or reaction to the chemo.
- Before going home, you'll get instructions for home care and any prescriptions you might need.

How often and for how long will I need chemo?

How often you get chemo and how long your treatment lasts will depend on:

- The kind of cancer you have
- The goals of the treatment
- The types of chemo being used
- The way the chemo is given
- How your body responds to treatment

You may get treatments daily, weekly, or monthly, but they're usually given in on-and-off cycles. For example, on a 3-week cycle, you might get chemo the first 2 weeks and then have a week off. The time off lets your body build healthy new cells and regain its strength. Ask your cancer care team how many cycles are planned and how long they expect your treatment to last.

How long does chemo stay in the body?

Your kidneys and liver break down most chemo medicines which then leave your body through urine or stool. How long it takes your body to get rid of the chemo depends on many things, including:

- The type of chemo
- Other medicines you take
- Your age
- How well your kidneys and liver work

Your cancer care team will tell you if you need to take any special precautions because of the chemo you are getting. To learn more, see [Chemotherapy Safety](#).

Hyperlinks

1. www.cancer.org/cancer/supportive-care/iv-lines-ports.html
2. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/hair-loss/cold-caps.html
3. www.cancer.org/support-programs-and-services/road-to-recovery.html
4. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/hair-loss/choosing-and-wearing-wig.html
5. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/hair-loss/how-to-wear-a-scarf.html
6. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/hair-loss/how-to-wear-a-hat.html

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Getting Oral Chemotherapy

Many types of chemotherapy (chemo) are given by infusion or injection, but some are given by mouth. Oral chemo is just as strong as chemo given by infusion or injection and can have the same side effects.

What is oral chemo?

Which cancers are treated with oral chemo?

How often will I need to take oral chemotherapy?

How and when to take oral chemo

Will I still need to see my doctor if taking oral chemo?

Special handling for oral chemo

How long does oral chemo stay in the body?

What are the side effects of oral chemo?

Questions to ask before starting oral chemo

Oral chemo costs

What is oral chemo?

Oral chemo is chemo that can be taken by mouth as pills, capsules, or liquids. Like other types of chemo, oral chemo may be given in rounds or cycles. This cuts down on the harm to healthy cells and allows them to recover from the side effects of the chemo. However, some types of oral chemo are given continuously, requiring close monitoring of side effects.

Examples of oral chemo include capecitabine, cyclophosphamide, and etoposide.

Which cancers are treated with oral chemo?

With more types of oral chemo now available, it's being used more often than ever. But it isn't an option for every type of cancer, and it may not be right for everyone. Cancers often treated with oral chemo include some forms of:

- Breast cancer
- Colorectal cancer
- Small cell lung cancer
- Leukemia
- Lymphoma
- Multiple myeloma

Oral chemo may be used for other types of cancer as well.

How often will I need to take oral chemotherapy?

Oral chemo is often given in cycles, just like chemo infusions. You may need treatments daily (sometimes multiple times a day), weekly, or monthly, but they're usually given in on-and-off cycles. This means, for example, that you may take chemo the first 2 weeks and then have a week off, making it a cycle that will start over every 3 weeks. The time off lets your body build healthy new cells and regain its strength.

Sometimes, oral chemo is given continuously, without breaks or time off. This keeps a steady level of medicine in your blood and gives cancer cells constant exposure to the chemo as they grow and divide.

Your cancer care team will tell you how many cycles are planned and how long they expect your treatment to last. How often you'll need to take oral chemo and how long your treatment lasts will depend on:

- The kind of cancer you have
- The goals of the treatment
- The types of chemo being used
- How your body responds to them

How and when to take oral chemo

The success of oral chemo taken at home depends on you – it's important to take the right dose of the chemo, at the right time, just as you've been told. Your cancer care team is there to help you do this.

Oral chemo doses are designed to keep a steady level of the chemo in your body to kill the cancer cells. Not taking your chemo the right way can affect how well it works. Your care team should give you clear instructions on how much and when to take your oral chemo. **You need to take the exact dosage, at the exact right time, for exactly as long as you're supposed to do so.** Sometimes dose changes are needed, but don't make any changes unless your doctor tells you to do so.

If you miss a dose or are late taking one, tell your doctor or nurse about it. They need to know about this when deciding if treatment is working. It may also help the doctor decide whether to change how much of the medicine you take or when you take it.

Sometimes there are food and drug interactions that can change how well the chemo works, making it less effective or causing additional side effects. Make sure you know:

- If you need to avoid any other medicines, food, vitamins, herbs, supplements, skin lotions, or other treatments before you begin oral chemo
- If the tablets or capsules can be opened, broken, or crushed
- If you can mix them with food or liquid to take them

Will I still need to see my doctor if taking oral chemo?

Even though you take oral chemo at home, you will still need to see your cancer care team. They will:

- Watch for changes in the cancer
- Check how you are doing with your chemo plan
- Do [blood tests](#)¹ and [scans](#)² to see how your body and the cancer are doing with the chemo.

Special handling for oral chemo

Chemo you swallow is just as strong as other forms of chemo. Many are also considered hazardous. There are usually special precautions for storing and handling oral chemo. For example:

- You might be told to be careful not to let others come into contact with the chemo or your body fluids while taking it and for a time after taking it.
- Sometimes you need to wear gloves when touching the pills or capsules.
- Some chemo has to be kept in the bottle or box they came in.
- Some types of chemo and the packages they come in need to be disposed of in a certain way.
- Some might have to be taken back to the pharmacy to be thrown away safely.

To learn more, see [Chemotherapy Safety](#).

How long does oral chemo stay in the body?

Many people wonder how long the actual chemo stays in their body and how it's removed. Your kidneys and liver break down most chemo medicines which then leave your body through urine or stool. How long it takes your body to get rid of the chemo depends on many things, including the type of chemo you get, other medicines you take, your age, and how well your kidneys and liver work. Your cancer care team will tell you if you need to take any special precautions because of the chemo you are taking.

What are the side effects of oral chemo?

The [side effects](#)³ of oral chemo vary from medicine to medicine and from person to person. Your cancer care team may not be able to tell you what side effects you'll have, but they can tell you what to watch for.

If you do have side effects, tell your team about them as soon as they happen, so they can help you manage them. Your doctor may change how much you take or give you other medicines to help you feel better.

If you aren't sure about a side effect and can't reach your doctor, don't take your chemo until you talk to someone on your cancer care team.

Taking chemo at home gives you more freedom without having a lot of treatment visits. You may not be seeing your doctor and nurses very often but be sure to call them with any questions or concerns you have.

Questions to ask before starting oral chemo

Be sure to talk to your care team about what steps to take if you have problems with your medication, including:

- What to do if you miss a dose
- What to do if you lose or misplace a pill

- What to do if you have trouble swallowing, [throw up, or feel sick to your stomach](#)⁴
- How to reach your care team if you have trouble late at night or on the weekend

Oral chemo costs

Oral chemo can be expensive. Make sure you ask your doctor about the cost of your treatment so you are not surprised when you get to the pharmacy or when you get your bill.

Depending on the type of chemo, some insurances don't cover the full cost or may not cover it at all. Sometimes you can get assistance, but many people have to pay more of their own money for oral chemo than they would for chemo infusions. If you have [health insurance](#)⁵, this might mean a higher co-pay. Make sure you know how much you'll have to pay for each treatment. You can call the American Cancer Society at 1-800-227-2345 for more information about financial assistance.

Hyperlinks

1. www.cancer.org/cancer/diagnosis-staging/tests/understanding-your-lab-test-results.html
2. www.cancer.org/cancer/diagnosis-staging/tests/imaging-tests/imaging-radiology-tests-for-cancer.html
3. www.cancer.org/cancer/managing-cancer/side-effects.html
4. www.cancer.org/cancer/managing-cancer/side-effects/eating-problems/nausea-and-vomiting.html
5. www.cancer.org/cancer/financial-insurance-matters/understanding-health-insurance.html

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Getting Topical Chemotherapy

While many types of chemotherapy (chemo) are given by infusion or injection, there are also types of chemo that are rubbed on the skin. Usually, these come in a cream or ointment and are applied directly to the site of the cancer.

[What is topical chemo?](#)

[How often will I need topical chemotherapy?](#)

[How and when to use topical chemo](#)

[Will I still need to see my doctor if using topical chemo?](#)

[Special handling for topical chemo](#)

[How long does topical chemo stay in the body?](#)

[What are the side effects of topical chemo?](#)

[Questions to ask before starting topical chemo](#)

What is topical chemo?

Topical chemo is chemo that comes in a cream or ointment that is rubbed onto the skin. It is used most commonly as a [non-surgical treatment](#)¹ for certain types of skin

cancer, such as basal or squamous cell, or precancerous changes, such as actinic keratoses. Because the chemo is only applied to the skin, it doesn't spread throughout the body, so it doesn't cause the same side effects as [systemic chemotherapy](#)² (treatment that affects the whole body). Examples of topical chemo include 5-fluorouracil (5-FU) and tirbanibulin (Klisyri).

How often will I need topical chemotherapy?

You may need topical chemo treatments once or twice daily for a period of days to weeks. Your cancer care team or dermatologist will tell you how often and for how long to apply the topical chemo.

How often you use topical chemo and how long your treatment lasts depend on:

- The kind of cancer
- The goals of the treatment
- The chemo being used
- How your body responds to it

How and when to use topical chemo

Most often, topical chemotherapy comes as a cream or ointment from the pharmacy, and you apply it at home. Your doctor or pharmacist will give you specific instructions for how much to use, where on your skin to apply it, and how often. Be sure to follow the instructions on the label about how to use the medication. Wash your hands before and after use.

The success of topical chemo taken at home depends on you—it's important to take the right dose at the right time, just as you've been told. Your cancer care team is there to help you do this.

If you miss a dose or are late taking one, tell your doctor or nurse about it. They need to know about this when deciding if treatment is working. It may also help the doctor decide whether to change how much of the medicine you take or when you take it.

Will I still need to see my doctor if using topical chemo?

Even though you apply topical chemo at home, you will still need to see your cancer care team. They will:

- Watch for changes in the cancer
- Monitor you for side effects
- See how you are doing with your chemo plan

Special handling for topical chemo

Chemo you apply to your skin is just as strong as other forms of chemo. Many are also considered hazardous. There are usually special precautions for storing and handling topical chemo. For example:

- You might be told not to let others come into contact with the chemo or your body fluids while taking it and for a time after taking it.
- Wear gloves or use applicators to apply the chemo and avoid touching the area afterward.
- Some types of chemo and the packages they come in need to be disposed of in a certain way.
- Some might need to be returned to the pharmacy to be thrown away safely.

To learn more, see [Chemotherapy Safety](#).

How long does topical chemo stay in the body?

Many people wonder how long the actual chemo stays in their body and how it's removed. Your kidneys and liver break down most chemo medicines, which then leave your body through urine or stool. How long it takes your body to get rid of the chemo depends on many things, including:

- The type of chemo
- Other medicines you take
- Your age
- How well your kidneys and liver work

Your cancer care team will tell you if you need to take any special precautions because of the chemo you are taking.

What are the side effects of topical chemo?

The [side effects](#)³ of topical chemo vary from medicine to medicine and from person to person. Your cancer care team may not be able to tell you what side effects you'll have, but they can tell you what to watch for. In general:

- You may experience itching, dryness, swelling, or scarring in the treated area.
- Your skin will likely be reddened and very sensitive for a few weeks.
- Some topical chemo can make the skin more sensitive to sunlight, so treated areas must be protected from the sun to prevent sunburn for a few weeks after treatment.

Tell your cancer care team about side effects right away so they can help you manage any problems. Your doctor may change how much you use or give you other medicines to help you feel better. If you aren't sure about a side effect and can't reach your doctor, don't apply your chemo until you talk to someone on your cancer care team.

Applying chemo topically at home gives you more freedom without having a lot of treatment visits. You may not be seeing your doctor and nurses very often, but be sure to call them with any questions or concerns you have.

Questions to ask before starting topical chemo

Be sure to talk to your care team about what steps to take if you have problems with your medication, including:

- What to do if you miss a dose
- What to do if you lose or misplace the medication
- What to do if you have trouble applying the medication or experience side effects
- How to reach your care team if you have trouble late at night or on the weekend

Hyperlinks

1. www.cancer.org/cancer/types/basal-and-squamous-cell-skin-cancer/treating/other-than-surgery.html
2. www.cancer.org/cancer/types/basal-and-squamous-cell-skin-cancer/treating/systemic-chemotherapy.html
3. www.cancer.org/cancer/managing-cancer/side-effects.html

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Chemotherapy Side Effects

Chemotherapy (chemo) treats many types of cancer effectively. But like other cancer treatments, it often causes side effects. It is important to be aware of possible side effects from chemo so you know what to watch for and can talk with your cancer care team about preventing and relieving them.

The side effects of chemo are different for each person. They depend on the type of cancer, location, types of chemo and dose, and your general health.

[Why does chemo cause side effects?](#)

[What do I need to know about side effects?](#)

[How long do chemo side effects last?](#)

[What are common side effects of chemo?](#)

[Chemo drug interactions and side effects](#)

[How vitamins affect chemo](#)

[When to call your cancer care team about chemo side effects](#)

The information below describes side effects that might be caused by traditional or standard chemotherapy. There are also other medicines that are used to treat cancer in different ways, including [targeted therapy](#)¹, [hormone therapy](#)² and [immunotherapy](#)³.

Why does chemo cause side effects?

Cancer cells grow fast, and chemo kills fast-growing cells. But because chemo travels throughout the entire body, it can affect normal, healthy cells that are fast-growing, too. Damage to healthy cells causes side effects. Side effects are not always as bad as you might expect, but it's normal to worry about this part of cancer treatment.

The normal cells most likely to be damaged by chemo are:

- Blood-forming cells in the bone marrow
- Hair follicles
- Cells in the mouth, digestive tract, and reproductive system

Some types of chemo can damage cells in the heart, kidneys, bladder, lungs, and nervous system.

Sometimes, you can take medicines with the chemo to help protect your body's normal cells. There are also treatments to help relieve side effects.

Doctors try to give chemo at levels high enough to treat cancer while keeping side effects at a minimum. They also try to avoid using multiple types of chemo that have similar side effects.

What do I need to know about side effects?

- Every person doesn't get every possible side effect, and some people get few or none.
- The severity of side effects (how bad they are) varies greatly from person to person.
- Your doctor may give you medicines to help prevent certain side effects before they happen.
- Some chemo causes long-term side effects, like heart or nerve damage or fertility problems. Still, many people have no long-term problems from chemo. Ask your doctor if the chemo you're getting has long-term effects.

While side effects can be unpleasant, they must be weighed against the need to kill the cancer cells.

Be sure to talk to your cancer care team about:

- Which side effects are most common with your chemo
- How long they might last
- How severe they might be
- When and who to call if you have side effects

How long do chemo side effects last?

Many side effects are **short-term**. They go away fairly quickly after a chemo cycle, or they might last for as long as you are in treatment. But some might be **long-term** or **chronic side effects**. They start during treatment but continue even after treatment is over, sometimes lasting for months or years. Sometimes the side effects can last a lifetime, such as when chemo causes long-term damage to the heart, lungs, kidneys, or reproductive organs.

Certain types of chemo sometimes cause **late side effects** that start after treatment ends. These side effects can happen long after cancer treatment, such as a [second cancer](#)⁴ that may show up many years later.

People often become discouraged about how long their treatment lasts or the side effects they have. If you feel this way, talk to your cancer care team. They may be able to change your medicine or treatment schedule. Or they may suggest ways to reduce any pain and discomfort you have.

What are common side effects of chemo?

Side effects of chemo can be very different depending on the type you are getting. Here are some of the more common side effects caused by chemo:

- [Fatigue](#)⁵
- [Hair loss](#)⁶
- [Thrombocytopenia](#)⁷ (low platelet count)
- [Neutropenia](#)⁸ (low white blood counts)
- Increased risk of [infection](#)⁹
- [Anemia](#)¹⁰ (low red blood cell counts)
- [Nausea and vomiting](#)¹¹
- [Appetite changes](#)¹²
- [Constipation](#)¹³
- [Diarrhea](#)¹⁴
- [Mouth sores](#)¹⁵
- [Peripheral neuropathy](#)¹⁶ or other nerve problems
- [Skin](#)¹⁷ and [nail](#)¹⁸ changes such as rash, dry skin and color change
- [Bowel](#)¹⁹ or [bladder](#)²⁰ incontinence
- [Weight changes](#)²¹
- [Chemo brain](#)²², which can affect concentration and focus
- [Anxiety](#)²³
- [Depression](#)²⁴

- [Changes in libido and sexual function](#)²⁵
- [Fertility problems](#)²⁶

Learn more about these and other problems in [Managing Cancer-related Side Effects](#)²⁷.

Chemo drug interactions and side effects

When looking at how best to combine types of chemo, doctors must look at interactions between chemo drugs and other medicines the person is taking, including over-the-counter medicines, vitamins, and supplements. These interactions may make side effects worse and affect how well chemo works.

It's important that you tell your doctor about all medicines, including over-the-counter medicines, vitamins, and herbal or [dietary supplements](#)²⁸ you are taking—even if you only take them “as needed.”

For instance, platelets help blood clot and prevent bleeding. Many types of chemo lower the number of platelets for a time. Taking aspirin or other related medicines can also weaken blood platelets. This isn't a problem for healthy people with normal platelet counts, but if a person has low platelet counts from chemo, this combination might put them at risk of a serious bleeding problem.

Your doctor can talk with you about the safety of using other medicines, vitamins, and supplements while you are being treated for cancer.

How vitamins affect chemo

Many people want to take an active role in improving their overall health. They want to help their body's natural defenses fight the cancer and speed up their recovery from chemo. Most people think of vitamins as a safe way to improve health, so it's not surprising that many people with cancer take high doses of one or more vitamins. But some vitamins might make chemo less effective.

More research is needed, but until more is known about the effects of vitamins on chemo, keep these points in mind:

- Unless your doctor tells you to take certain vitamins, it's best not to take any.

- Always check with your doctor first before starting to take a vitamin of any kind, even a simple multivitamin.
- Ask your doctors if and when it might be OK to start taking vitamins after treatment.
- If you're concerned about nutrition, you can usually get plenty of vitamins by eating a well-balanced diet. See [Nutrition During and After Cancer Treatment](#)²⁹ to learn more.

When to call your cancer care team about chemo side effects

Because your cancer care team will give you lots of information about side effects, you might be more aware of physical changes. Do not take any physical symptoms you have lightly. Some side effects are short-lived and minor, but others may be a sign of serious problems. **Make sure you know how to reach someone on your team any time, including after hours, weekends, and holidays.**

Contact your cancer care team right away if you have any of the following symptoms:

- A fever higher than what your cancer care team has instructed (usually 100.5°F–101°F or higher, taken by mouth)
- Bleeding or unexplained bruising
- A rash
- An allergic reaction, such as swelling of the mouth or throat, severe itching, or trouble swallowing
- Intense chills
- Pain or soreness at the chemo injection site or catheter site
- Unusual pain, including intense headaches
- Shortness of breath or trouble breathing (If you're having trouble breathing, call 911 first.)
- Long-lasting diarrhea or vomiting
- Bloody stool or blood in your urine

Ask your cancer care team if there are any other problems they should know about right away.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/targeted-therapy.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/hormone-therapy.html
3. www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy.html
4. www.cancer.org/cancer/survivorship/long-term-health-concerns/second-cancers-in-adults.html
5. www.cancer.org/cancer/managing-cancer/side-effects/fatigue-weakness-sleep/fatigue.html
6. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails/hair-loss/coping-with-hair-loss.html
7. www.cancer.org/cancer/managing-cancer/side-effects/low-blood-counts/bleeding.html
8. www.cancer.org/cancer/managing-cancer/side-effects/low-blood-counts/neutropenia.html
9. www.cancer.org/cancer/managing-cancer/side-effects/infections.html
10. www.cancer.org/cancer/managing-cancer/side-effects/low-blood-counts/anemia.html
11. www.cancer.org/cancer/managing-cancer/side-effects/eating-problems/nausea-and-vomiting.html
12. www.cancer.org/cancer/managing-cancer/side-effects/eating-problems/poor-appetite.html
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22. www.cancer.org/cancer/managing-cancer/side-effects/changes-in-mood-or-thinking/chemo-brain.html
23. www.cancer.org/cancer/managing-cancer/side-effects/emotional-mood-changes/anxiety.html
24. www.cancer.org/cancer/managing-cancer/side-effects/emotional-mood-changes/depression.html
25. www.cancer.org/cancer/managing-cancer/side-effects/sexual-side-effects.html
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29. www.cancer.org/cancer/supportive-care/nutrition-activity-with-cancer.html

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Chemotherapy Safety

Chemotherapy (chemo) works by destroying cancer cells. Because it may also damage healthy cells, it can be hazardous to people who are exposed to it. This is why there are safety precautions for people who handle chemo.

[Precautions the cancer care team will take](#)

[Precautions when getting IV chemo at home](#)

[Precautions when taking chemo by mouth](#)

[Can I be around family and friends?](#)

[What to do during – and for at least 48 hours after – chemo](#)

Not all medicines and drugs to treat cancer work the same way or have the same safety precautions. The information below describes some safety concerns of traditional or standard chemotherapy. There are also other drugs that are used to treat cancer in different ways, including [targeted therapy](#)¹, [hormone therapy](#)², and [immunotherapy](#)³.

Precautions the cancer care team will take

You may notice your cancer care team using special clothing and protective equipment during and after your chemo treatments. This is to help keep them safe and prevent accidental exposure. These may include:

- Multiple pairs of gloves
- Gowns
- Goggles or plastic face shields
- Special tubing
- Plastic-backed pads

Precautions when getting IV chemo at home

Sometimes, chemo can be given at home using special equipment. This requires precautions for storing, handling, and receiving the chemo. Steps to take may include:

- Storing chemo in a safe place away from the reach of children or pets
- Wearing gloves when touching the chemo bag or pump
- Keeping a chemo spill kit in a safe place with instructions on how to use in the event of a spill.
- Knowing how to properly dispose of hazardous waste

If you are getting IV chemo at home, talk to your cancer care team about any special precautions needed. To learn more, see [Getting IV or Injectable Chemotherapy](#).

Precautions when taking chemo by mouth

Oral chemo (chemo you take by mouth) is usually taken at home. These types of chemo are as strong as other forms of chemo, and many are considered hazardous. There are usually special precautions for storing and handling oral chemo.

You might be told to prevent others from coming into contact with the chemo or your body fluids while taking it and for a time afterwards. Sometimes you need to wear gloves when touching the pills or capsules. Some oral chemo needs to be stored and disposed of in a certain way.

If you are taking an oral chemo, talk to your cancer care team about any special precautions needed at home. To learn more, see [Getting Oral Chemotherapy](#).

Can I be around family and friends?

Unless your care team tells you differently, you can usually be around family and friends while you're getting chemo. On treatment days, family and friends can often come with you. However, some treatment centers only allow patients in the infusion area and visitors may need to stay in the waiting room. Also, most treatment centers do not allow children.

Ask your cancer care team if there are any special precautions you or your loved ones need to follow.

What to do during – and for at least 48 hours after – chemo

It generally takes at least 48 hours for your body to break down and/or get rid of most chemo. Some chemo passes more quickly while others take longer to leave your body. Talk to your doctor or nurse about how the chemo you are getting is passed and how long it may take.

Most of the chemo waste comes out in your body fluids, such as your urine and stool. The waste is also found in your blood and sometimes in vomit, saliva, sweat, semen, and vaginal fluids. Chemo and its waste can harm or irritate skin, and may cause damage. Other people and pets could be exposed to chemo waste for a few days if they come into contact with any of your body fluids.

Here are things you can do to help keep your family, visitors, and pets safe during this time:

When using the bathroom

- If possible, have others use a separate toilet during this time. If this isn't possible, clean the toilet seat after each time you use it with a sanitizing wipe.
- Put the lid down before flushing to avoid splashing. Flush the toilet twice after you use it, if you have a low volume flush toilet.
- Sit on the toilet when you use it to cut down on splashing.
- Always wash your hands with warm water and soap after using the toilet.
- Keep the toilet lid down when you're not using it to keep pets from drinking the water.

If you vomit

- If you vomit into the toilet, clean off all splashes along the seat and rim using a sanitizing wipe, then flush.
- If you vomit into a bucket or basin, carefully empty it into the toilet without splashing the contents. Wash out the bucket with hot, soapy water and rinse it. Empty the wash and rinse water into the toilet, then flush.

If others might come into contact with body fluids

- If others need to touch any of your body fluids or something that has been in contact with your body fluids (like a bucket or basin), they should wear 2 pairs of throw-away gloves. These gloves can be bought in most pharmacies. They should always wash their hands with warm water and soap afterward – even if they had gloves on.
- If others come in contact with any of your body fluids, they should wash the area very well with warm water and soap. It's not likely to cause any harm but try to avoid this. At your next visit, let your doctor know this happened. Being exposed often may lead to problems, and extra care should be taken to avoid this.

Laundry safety

- Any clothes or sheets that have body fluids on them should be washed in your washing machine – not by hand. Wash them twice in hot water with regular laundry detergent. Do not wash them with other clothes. If they can't be washed right away, seal them in a plastic bag until they can be washed.
- If using throw-away adult diapers, underwear, or sanitary pads, seal them in a leak-proof, plastic bag and throw them away with your regular trash.

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/targeted-therapy.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/hormone-therapy.html
3. www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy.html

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